

Commit 32ee47e

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ActuallyNull committed on Aug 10

added a bandpass filter, smote and signal resampling, and improved preprocess group function

main 1 parent e0c92f0 commit 32ee47e

- data_processing.py
- testingScripts
- ecg_processor_test.py

2 files changed +32 -12 lines changed Search within code

```
data_processing.py +29 -9
... @@ -1,8 +1,11 @@
1 - from sklearn.model_selection import train_test_split
2 1 import wfdb
3 2 import numpy as np
4 3 import os
5 4 import pandas as pd
6 5 + from sklearn.preprocessing import MinMaxScaler
7 6 + from scipy.signal import butter, filtfilt, resample
8 7 + from imblearn.over_sampling import SMOTE
9 8 + from collections import Counter
10
11 dataset_folders = ["mit-bih", "training2017/training2017"]
12 records = []
13
14 @@ -13,14 +16,26 @@
15
16 X = []
17 y = []
18
19 + def butter_bandpass_filter(signal, fs, low_pass=0.5, high_pass=50, order=5):
20 +     nyq = 0.5 * fs
21 +     low = low_pass / nyq
22 +     high = high_pass / nyq
23 +     b,a = butter(order, [low, high], btype='band')
24 +     return filtfilt(b, a, signal)
25 +
26 + def smote_resampling(X, y):
27 +     smote = SMOTE(random_state=42)
28 +     X_resampled, y_resampled = smote.fit_resample(X, y)
29 +     return X_resampled, y_resampled
30 +
31 + def signal_resampling(signal, target_length):
32 +     resample(signal, target_length)
33 +
34 def standardise_dataset(X):
35     X_scaled = np.array([(x - np.mean(x)) / np.std(x) for x in X])
36     return X_scaled
37
38 - from sklearn.preprocessing import MinMaxScaler
39 -
40 def normalise_dataset(X):
41     # Reshape X to 2D if needed (n_samples, n_features)
42
43     scaler = MinMaxScaler()
44     X_norm = scaler.fit_transform(X)
45     return X_norm
46
47 @@ -63,12 +78,17 @@ def load_physionet_dataset(physionet_path, reference_file_path, X, y):
48
49     y = np.array(y)
50     return X, y
51
52 - def process_physionet_dataset(physionet_path, reference_file_path, X, y): #X and y should be empty lists
53 + def preprocess_dataset(physionet_path, reference_file_path, X, y, fs=300): #X and y should be empty lists
54
55     X, y = load_physionet_dataset(physionet_path, reference_file_path, X, y)
56
57     - X = standardise_dataset(X)
58     - X = normalise_dataset(X)
59     - return X, y
60
61 + X_filtered = np.array([butter_bandpass_filter(x, fs=fs) for x in X])
62 +
63 + #X_standardised = standardise_dataset(X_filtered)
64 + X_processed = normalise_dataset(X_filtered) # since this should be a dcnn, no standardisation needed
65 + y_processed = y
66
67
68 - X, y = process_physionet_dataset("training2017/training2017", "training2017/training2017/REFERENCE.csv", X, y)
69
70 + # Apply SMOTE to balance the dataset
71 + X_balanced, y_balanced = smote_resampling(X_processed, y_processed)
72 + return X_balanced, y_balanced
73
74
75 + X_processed, y_processed = preprocess_dataset("training2017/training2017", "training2017/training2017/REFERENCE.csv", X,
76 y)
77
78
```

```
testingScripts/ecg_processor_test.py +3 -3
@@ -3,12 +3,12 @@
3 3 import numpy as np
4 4 import pandas as pd
5 5
6 - record = wfdb.rdrecord("mit-bih/100")
7 + record = wfdb.rdrecord("training2017/training2017/A00001")
8 7 fs = record.fs
9 - ecg_signal = record.p_signal[:, 0]
10 + #ecg_signal = record.p_signal[:, 0]
11 9 duration = 5
12 10 samples = fs * duration
13 - print(record.comments)
14 + print(fs)
15
16
17 #_, rpeaks = nk.ecg_peaks(ecg_signal, sampling_rate=fs)
18 #plot = nk.events_plot(rpeaks['ECG_R_Peaks'][:5], ecg_signal[:6*fs])
19
20
```

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